

Pricing American Put Options with Least-Squares Monte Carlo

Longstaff–Schwartz vs. Tsitsiklis–Van Roy under a shared benchmark design

Ethan Yao

Why this project matters

- American option pricing is an **optimal stopping** problem.
- Need both:
 - a price,
 - an exercise rule.
- Direct trees / PDE methods work well in **simple 1D settings**.
- Richer contracts quickly create problems with:
 - larger state spaces,
 - path dependence,
 - changed dynamics.
- This project studies two regression-based simulation methods under **one common design**.

Goal: compare methodology, not mismatched case studies.

Methods compared

Longstaff–Schwartz (LSM)

- Regress **discounted realized future cash flows**
- Builds a stopping rule from estimated continuation values
- Main practical pricing method in this project

Tsitsiklis–Van Roy (TVR)

- Regress **discounted next-step estimated values**
- Fitted value iteration / approximate dynamic programming
- Matched comparator under the same testbeds

$$V_i(s) = \max\{h(s), C_i(s)\}$$

Key difference: same Bellman problem, different recursion target.

Experimental design: one benchmark layer, two extensions

- **Main validation layer:** vanilla American put under risk-neutral GBM
- **Shared inputs:**
 - $K = 40, r = 0.06$
 - $S_0 \in \{36, 38, 40, 42, 44\}$
 - $\sigma \in \{0.20, 0.40\}$
 - $T \in \{1, 2\}$
 - 50 exercise dates / year
 - 100,000 paths
- **Extensions:**
 - **Asian-style:** richer state vector (S_t, A_t)
 - **Jump-style:** changed simulator dynamics

Design rule: vanilla is the benchmark exercise; Asian and jump are structural extensions.

Benchmark hierarchy and implementation workflow

European BS lower reference → Direct American benchmarks (FD, CRR) →

Longstaff / TVR → Robustness + extensions

- Shared pieces wherever possible:
 - simulator,
 - basis construction,
 - benchmark logic.
- Different pieces:
 - **LSM**: continuation-cash-flow regression,
 - **TVR**: fitted value recursion.
- In both methods, the fitted policy is then **evaluated on fresh paths**.

Why this matters: same environment, different recursion logic.

Vanilla benchmark: methods are sensible, but not interchangeable

Case	FD	LSMC (OOS)	TVR (OOS)	Pattern
$S_0 = 36, \sigma = 0.2, T = 1$	4.463	4.206	4.385	TVR closer
$S_0 = 40, \sigma = 0.2, T = 1$	2.299	2.444	2.278	LSM high, TVR close
$S_0 = 40, \sigma = 0.2, T = 2$	2.873	2.727	2.829	both reasonable
$S_0 = 40, \sigma = 0.4, T = 2$	6.897	6.971	6.859	both close
$S_0 = 44, \sigma = 0.4, T = 1$	3.924	4.132	3.838	LSM high, TVR low

- Both methods produce **economically sensible American put values**.
- LSM can price **above or below** the FD benchmark.
- TVR tends to stay **near FD** and often slightly **below** it.

Takeaway: differences are methodological, not noise from different setups.

Vanilla visuals: benchmark alignment

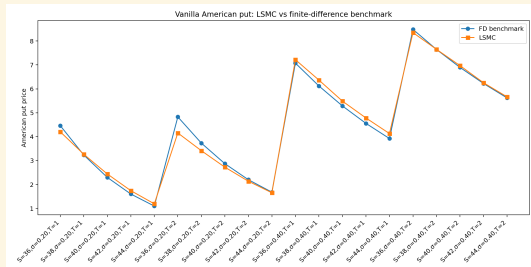


Figure 1: Longstaff benchmark vs finite difference.

LSM: tighter than same-path pricing, but still case-dependent around FD.

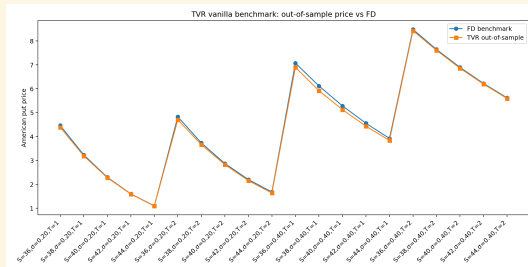


Figure 2: TVR benchmark vs finite difference.

TVR: clusters more closely around FD across the shared grid.

Validation matters: same-path optimism is mainly a Longstaff issue

Representative case	In-sample	Out-of-sample	Fit - Eval
LSM: $S_0 = 36$, $\sigma = 0.2$, $T = 1$	4.452	4.206	0.247
LSM: $S_0 = 40$, $\sigma = 0.2$, $T = 2$	3.060	2.739	0.321
LSM: $S_0 = 40$, $\sigma = 0.4$, $T = 2$	7.810	6.956	0.854
TVR: $S_0 = 36$, $\sigma = 0.2$, $T = 1$	4.386	4.381	0.005
TVR: $S_0 = 40$, $\sigma = 0.2$, $T = 2$	2.804	2.817	-0.014
TVR: $S_0 = 44$, $\sigma = 0.4$, $T = 1$	3.865	3.840	0.025

- LSM shows **clear same-path optimism**.
- TVR fit/eval gaps are **much smaller**.

Takeaway: out-of-sample policy evaluation is not optional.

Robustness: what moves the price?

Longstaff - Path count reduces SE, but not monotone benchmark error - Sensitive to exercise grid - Sensitive to basis choice - Seed variation is smaller than the main benchmark gaps

TVR - Also benefits from more paths - More stable across path budgets - More directional improvement with richer bases - Still mildly conservative against FD

Main message: price quality depends on more than path count: - basis design, - exercise grid, - validation structure, - approximation target.

Robustness snapshots

Test	Case	LSM pattern	TVR pattern
Paths	(40, 0.2, 1)	2.317 $\rightarrow$ 2.447	2.270 $\rightarrow$ 2.282
Paths	(40, 0.4, 2)	6.935 $\rightarrow$ 6.958	6.889 $\rightarrow$ 6.850
Grid	(40, 0.2, 1)	2.561 $\rightarrow$ 2.344	2.295 $\rightarrow$ 2.281
Grid	(40, 0.4, 2)	7.210 $\rightarrow$ 6.827	6.864 $\rightarrow$ 6.819
Basis	(40, 0.2, 1)	2.385 $\rightarrow$ 2.458	2.151 $\rightarrow$ 2.285
Basis	(40, 0.4, 2)	6.890 $\rightarrow$ 6.992	6.466 $\rightarrow$ 6.862

- LSM is more sensitive to the interaction of fit and stopping-rule construction.
- TVR is more constrained by **under-parameterization** when the basis is too small.

Asian-style extension: matched multistate study

- Contract: American–Bermuda–Asian put
- Shared design:
 - $K = 100$, $r = 0.06$, $\sigma = 0.20$, $T = 2$
 - lockout = 0.25
 - 100 exercise dates / year
- State is now **two-dimensional**:
 - current spot S_t
 - running arithmetic average A_t
- Immediate payoff depends on the average:

$$h(S_t, A_t) = (K - A_t)^+$$

Purpose: move from 1D vanilla validation to a richer continuation problem.

Asian-style results: both methods work, policies differ

Representative case	LSM (OOS)	LSM Fit-Eval	TVR (OOS)	TVR - LSM
$A_0 = 90, S_0 = 80$	13.777	1.016	17.498	3.720
$A_0 = 100, S_0 = 100$	3.213	0.122	3.598	0.384
$A_0 = 110, S_0 = 110$	1.066	0.068	1.078	0.013

- Both methods show the right comparative statics.
- LSM still has **positive fit-vs-eval gaps**.
- TVR is more **validation-stable** and often gives **higher values**.

Interpretation: multistate comparison only — not a direct PDE benchmark exercise.

Asian-style visuals: continuation behavior differs

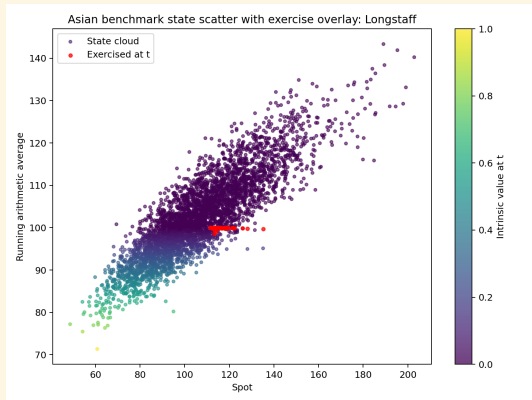


Figure 3: Longstaff Asian state cloud with exercise overlay.

LSM: more exercise-prone in borderline states.

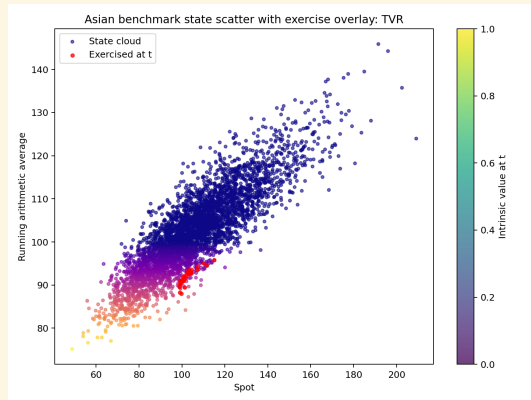


Figure 4: TVR Asian state cloud with exercise overlay.

TVR: more continuation-oriented under the same contract design.

Jump-style extension: matched changed-dynamics study

- Same vanilla American put payoff
- Change the **simulator**, not the workflow
- Shared setup:
 - $K = 40, r = 0.06, T = 1$
 - 26 exercise dates / year
 - 50,000 paths
 - $S_0 \in \{36, 38, 40, 42, 44\}$
- Two matched cases:
 - **No jump:** $\sigma = 0.30, \lambda = 0$
 - **Jump-to-ruin:** $\sigma = 0.20, \lambda = 0.05$

Purpose: test whether both methods remain coherent when dynamics change.

Jump-style results: changed dynamics, same methodological contrast

Case at $S_0 = 40$	American	Euro MC	Premium	Fit-Eval
LSM no-jump	4.093	3.570	0.523	0.235
LSM jump-to-ruin	3.614	3.220	0.394	0.149
TVR no-jump	3.743	3.588	0.155	0.001
TVR jump-to-ruin	3.277	3.217	0.060	-0.020

- Values fall with spot in both model cases.
- American values remain above European MC values.
- In this matched design, **jump-to-ruin values are generally lower** than no-jump values.
- TVR remains more conservative than LSM.

Jump-style visuals: price ranking + exercised-state summary

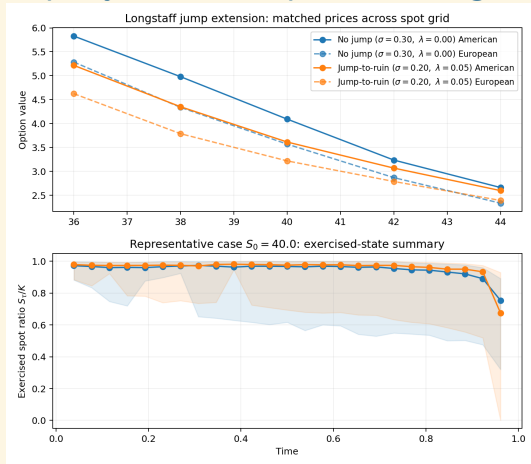


Figure 5: Longstaff matched jump extension: prices across spot grid and representative exercised-state summary.

LSM: coherent under changed dynamics

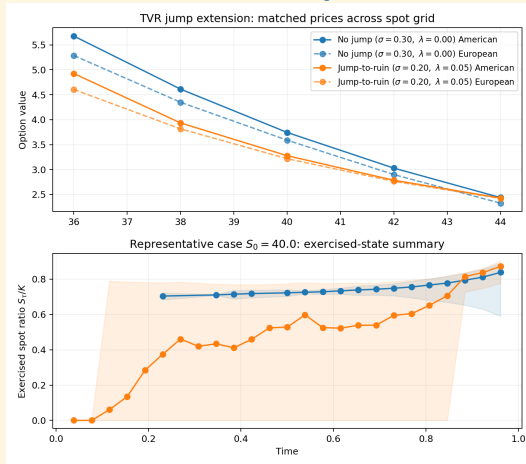


Figure 6: TVR matched jump extension: prices across spot grid and representative exercised-state summary.

TVR: same ranking, smaller fit/eval gap

Direct Longstaff vs TVR comparison

Longstaff–Schwartz - More directly tied to stopping-rule construction - Strong practical simulation method - More sensitive to: - fit/eval separation, - basis design, - exercise-grid design - Can look optimistic if only same-path values are reported

Tsitsiklis–Van Roy - More naturally read as fitted value iteration - More stable under matched out-of-sample evaluation - Often mildly conservative in benchmark alignment - More continuation-oriented in the extensions

Bottom line: same optimal stopping problem, different numerical behavior.

Implementation and computational testing

- The project does **not** claim strict numerical convergence proofs.
- Instead, it uses the papers' ideas to motivate a careful empirical workflow:
 - direct benchmarking where possible,
 - independent policy evaluation,
 - matched testbeds across methods,
 - robustness checks when approximation design matters.
- Vanilla is the main validation layer.
- Asian and jump are extension layers for:
 - richer state representation,
 - changed dynamics,
 - workflow flexibility.

Testing claim: not production pricing — but defensible implementation evidence.

Final takeaways

- Regression-based American pricing is **flexible and useful**.
- But the output depends on more than simulation size:
 - basis design,
 - state representation,
 - exercise grid,
 - validation structure.
- Longstaff and TVR are **not numerically interchangeable**.
- Under the shared design here:
 - **LSM** is more directly tied to practical stopping-rule construction,
 - **TVR** is more validation-stable and mildly conservative.
- Strongest project contribution:
 - one clean vanilla benchmark,
 - two matched extensions,
 - one coherent methodological comparison.

References

Longstaff, F. A., & Schwartz, E. S. (2001). *Valuing American options by simulation: A simple least-squares approach*.

Tsitsiklis, J. N., & Van Roy, B. (2001). *Regression methods for pricing complex American-style options*.